

Prediction of functional outcome of ischemic stroke patients in northwest China

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Abstract

Objectives: To identify the clinical factors which predicted the outcome of ischemic stroke patients in northwest China.

Patients and methods: We retrospectively reviewed 489 consecutive patients with ischemic stroke admitted to the Neurology Department of Xijing Hospital. Demographic, clinical and laboratory data were recorded. Follow-up assessments were performed by telephone interviews or letters. The clinical outcome was assessed by using the modified Rankin Scale (mRS) and categorized as good (score 0–2) or poor (score 3–6) outcomes. Univariate and multivariate logistic regression analyses were performed to explore predictors of ischemic stroke.

Results: The follow-up period was up to 47 months (mean, 28.3 months). Fifty-five patients (11.2%) were lost. Among these 434 patients, 244 (56.2%) patients had good outcome and 190 (43.8%) had poor outcome. The poor outcome was associated with old age (OR: 3.505; CI 95%: 2.100–5.849), lower educational level (OR: 0.686; CI 95%: 0.570–0.825), having stroke history (OR: 2.481; CI 95%: 1.442–4.268), and higher NIHSS total score (OR: 2.619; CI 95%: 1.584–4.330).

Conclusion: The results suggest that age, the educational level, stroke history, and NIHSS score are useful in the prediction of functional outcome of ischemic stroke in Chinese northwest area.

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Keywords: Ischemic stroke; Retrospective study; Prognosis

1. Introduction

Stroke is the third-leading cause of death in Western industrialized countries and a leading cause of chronic disability and suffering. In China, stroke is the second cause for mortality in all diseases. Particularly, ischemic strokes account for 60–80% of these strokes. With the rapid increase of aging population in China, stroke has become a critical cause for the health of the old.

In caring for stroke patients, clinicians and investigators often need to estimate the prognosis of stroke. Accurate prediction of acute stroke is important for several reasons. The role of clinicians in stroke management is not confined

to treatment, and the confidence of the patient and family can be greatly enhanced by the ability to offer an accurate prognosis. A reliable prognosis allows better planning for supportive care, more accurate information to be given to relatives and resources to be allocated in a more efficient way. Numerous clinical factors have been identified as potential predictors of outcome such as age [1], gender [2], severity of presenting clinical deficit [3] and laboratory data [4,5]. Some studies showed that race was also a predictor [6–8], while data concerning prognosis after stroke were very limited in mainland China. In addition, some studies did not distinguish hemorrhagic with ischemic stroke [9].

The purpose of the hospital-based study was to determine the clinical factors that predicted the outcome of ischemic stroke patients in northwest China.

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2. Patients and methods

2.1. Subjects

Medical records were retrieved by cross checking the hospital diagnosis data system. Study subjects were recruited from consecutive patients, who admitted to neurological department of XiJing Hospital with ischemic stroke between January 2002 and June 2005. The Xijing Hospital is an affiliated hospital of the Fourth Military Medical University within a metropolitan area of seven million inhabitants in Xi'an city, Shaanxi province. It is also the biggest general hospital with 1700 beds in northwest China. The Department of Neurology has 60 beds with 800–1200 inpatients per year coming from Shaanxi, Shanxi, Henan, Gansu and Qinghai provinces. The diagnosis of stroke was established according to World Health Organization criteria [10] by a neurologist and confirmed on CT, conventional MRI brain images, or both. Patients with hemorrhagic stroke, including subarachnoid hemorrhage, or reduced life expectancy (due to a severe concomitant illness, such as cancer, end-stage heart failure, liver failure) and TIA were excluded from the study. Patients with age below 18 years or time from onset to admission more than 14 days were also excluded.

2.2. Procedure

By review of the hospital records, demographic, previous history, clinical and laboratory data were collected with the use of a specially formulated questionnaire by the trained neurologists. Selection of variables was based on a systematic literature review for independent prognostic factors for outcome after ischemic stroke and these variables were commonly recorded in ischemic stroke patients of our department. Twenty-one variables were enrolled, including sex, age at the time of stroke, degree of education, smoking, previous history such as stroke, atrial fibrillation, hypertension and diabetes mellitus, body temperature, systolic pressure, diastolic pressure, leucocyte count, blood glucose, urea nitrogen, creatinine, fibrinogen, complication, Glasgow Coma Scale (GCS), NIHSS score, OCSP type, time from onset to admission. The degree of education was classified as illiteracy, elementary school, junior high school, senior high school, and university. Smoking was defined if smoking was more than 1 year noted in the medical records. The following comorbid conditions such as stroke, atrial fibrillation, hypertension and diabetes mellitus were noted positive if they were recorded as positive in the medical notes. Body temperature and pressure were taken from the earliest record within 6 h after admission and laboratory data were taken within 72 h. Neurologic impairment was measured at admission by using standardized scales, such as the Glasgow Coma Scale (GCS) [11] and the National Institute of Health Stroke Scale (NIHSS) [12]. The stroke types were classified according to the Oxfordshire Community Stroke Project (OCSP) as total anterior

circulation stroke (TACS), partial anterior circulation stroke (PACS), lacunar stroke (LACS), or posterior circulation stroke (POCS) [13]. Clinical assessment was performed with 5 days after admission. Medical complications in-hospital were defined by nonneurologic complications requiring intervention. These included the occurrence of pressure sore, urinary tract infection, chest infection, painful shoulder, pyrexial illness, deep vein thrombosis, depression, acute retention of urine, bleeding from the gastrointestinal tract, or pulmonary embolism. Case records with missing data were eliminated from the analysis. If patients had stroke more than once during the period, the data of the first stroke was used in the present analysis. To guarantee the validity of the registration, all available information of every patient was rechecked weekly by the study group composed of neurologists and epidemiologists.

2.3. Follow-up

All telephone interviewers were trained with content and method of interview. Clinical follow-up was performed strictly blinded to the patients' clinical features. If the patient could not be reached via telephone, a follow-up letter was sent. If still no information of the patient outcome could be obtained, the patient was defined lost. Each patient's disability was assessed according to the modified Rankin Scale (mRS) and categorized as good outcome (score 0–2) or poor outcome (score 3–6). After 2 weeks from follow-up, 10% patients from sampling were rechecked by another researcher who was blind to the patients' clinical features and the first follow-up result. The research group discussed if two follow-up visits had discrepancy in categorizing patients' disability.

2.4. Statistical analysis

All collected data were checked by two physicians for completeness and plausibility, and then were entered in duplicate into the database with Epi Data 2.0 by trained personnel. The reasonable interval of data was established and the consistency of two personnel was also checked during the period of entering. For each numeric variable, the normality of distribution was preliminarily assessed by the Kolmogorov–Smirnov test. These variables were generally expressed as mean (S.D.). However, the variables were expressed median (25th quartile, 75th quartile, IQR) when their distributions were not normal. All qualitative data were presented as numbers and percentages. Baseline difference between the follow-up group and the lost group was examined by means of the χ^2 -test or the Mann–Whitney *U*-test. The relation of each variable to outcome categories was separately tested by the two-sample *t*-test, the χ^2 -test, or the Mann–Whitney *U*-test as appropriate. Predictor variables that showed a significant relation with the outcome variables using a liberal significance level of $p < 0.10$ were selected for the subsequent development of the multivariate regression

model. This selection increased the power for selection of true predictors. Then those numeric variables were individually categorized in accordance with those accepted in clinical practice and previous reports [14]. To determine which combination of predictor variables led to the best predictive model, multivariate logistic regression analysis were carried out by using stepwise forward (probability of F to enter = 0.05 and probability of F to remove = 0.10). Selection of variables in the multivariate models was based upon likelihood ratio tests as well as judgment of the odd ratios of the variables, and their theoretical plausibility. We determined the sensitivity and specificity of the model and assessed goodness of fit with the Hosmer and Lemeshow statistics. Coefficients with a p -value lower than 0.05 were considered as significant. The statistical analyses were performed by using the SPSS for Windows 10.0 statistical package. Ethical approval was not needed for this study because only medical records were used.

3. Results

3.1. Subjects

Seven hundred and eighty-one consecutively ischemic stroke patients were admitted to the Neurology Department of our Hospital during the 42 months period. Patients ($n = 292$) were excluded for the following reasons: time from onset to admission more than 14 days ($n = 161$), age less than 18 ($n = 8$), transient ischemic attacks ($n = 43$), cancer ($n = 12$), chronic liver or renal failure ($n = 5$), missing data ($n = 63$). Among these patients with certain missing data, there were 38 patients missing fibrinogen, 29 patients missing clinical scores, 8 patients missing urea nitrogen and creatinine, 14 patients missing prior history, and 7 patients missing educational level. The remaining 489 ischemic stroke patients were entered into the study. Three hundred and twenty-eight (67.1%) patients were men and 161 (32.9%) patients were women. The median age of patients was 65 years (range: 22–93; IQR: 56, 71). The median time from onset to admission was 3 days (range: 0–14 days; IQR: 1, 5). The median hospital stay was 14 days (range: 0–87 days; IQR: 11, 19). Baseline characteristics are summarized in Table 1. The follow-up information was not available for 55 patients (11.2%). There were no significant differences in baseline characteristics between the follow-up group and the lost group (Table 2). The remaining 434 patients were entered into analysis. Of these patients, 19 patients (4.4%) died during hospital stay because of ischemic stroke and 415 (95.6%) were followed up by telephone ($n = 387$) or letters ($n = 28$) with a mean of 22.98 months (range: 4–47 months; S.D.: 9.89 months). Among these follow-up patients, 19 (4.6%) died during the follow-up with a mean survival of 12.7 months and 396 (95.4%) were alive after a mean follow-up of 26.5 months. Fourteen (73.7%) died from a vascular cause (10 from recurrent fatal stroke, 4 from myocardial infarct) and

Table 1

Baseline characteristics of ischemic stroke patients ($n = 489$)

Characteristic	Value
Sex (male) ^a	328 (67.1)
Age (years) ^b	65 (56, 71)
Education ^a	
Illiteracy	44 (9.0)
Elementary school	84 (17.2)
Junior high school	122 (24.9)
Senior high school	128 (26.2)
University	111 (22.7)
Smoking ^a	233 (47.6)
Hypertension ^a	281 (57.5)
Diabetes mellitus ^a	99 (20.2)
Stroke history ^a	107 (21.9)
Atrial fibrillation ^a	52 (10.6)
Time from onset to admission (days) ^b	3 (1, 5)
Length of hospital stay (days) ^b	14 (11, 19)
Area ^a	
Shaanxi	284 (58.0)
Shanxi	83 (17.0)
Henan	61 (12.5)
Gansu	37 (7.6)
Ningxia	24 (4.9)

^a n (%).^b Median (IQR).

five (26.3%) from non-vascular causes. Among these 434 patients, 244 (56.2%) patients have good outcome and 190 (43.8%) have poor outcome. Forty-four patients randomly selected were rechecked in categorizing patients' disability independently by another researcher, and the κ -value was 0.847.

3.2. Univariate analysis of predictors of ischemic stroke (Table 3)

Table 3 shows the results of the univariate analysis. The following 17 factors were significantly associated with poor outcome: female ($p = 0.001$), old age ($p < 0.001$), lower education level ($p < 0.001$), history of previous stroke ($p = 0.001$)

Table 2

Patients' demographical and clinical characteristics between the follow-up group and the lost group

Variables	Follow-up group ($n = 434$)	Lost group ($n = 55$)	p
Gender (male) ^a	295 (68)	33 (60)	0.236
Age (>60 years) ^a	264 (60.8)	38 (69.1)	0.235
Education ^a (below university)	333 (76.7)	45 (81.8)	0.396
Smoking ^a	205 (47.2)	28 (50.9)	0.607
Hypertension ^a	252 (58.1)	29 (52.7)	0.451
Diabetes mellitus ^a	83 (19.1)	16 (29.1)	0.083
Stroke history ^a	98 (22.6)	9 (16.4)	0.293
Atrial fibrillation ^a	43 (9.9)	9 (16.4)	0.143
Area (Shannxi) ^a	258 (59.4)	26 (47.3)	0.085
NIH score ^b	4 (3, 6)	5 (2, 8)	0.552

^a n (%).^b Median (IQR).

Table 3

Univariate analysis of predictors of functional outcome of ischemic stroke ($n = 434$)

Variables	Poor ($n = 190$)	Good ($n = 244$)	Statistical	p -Value	OR (95% CI)
Gender (male) ^a	113 (59.5)	182 (74.6)	11.212	0.001*	0.50 (0.33–0.75)
Age (≤ 60) ^a	47 (24.7)	123 (50.4)	29.548	0.000*	0.32 (0.21–0.49)
Education ^a					
Illiteracy	25 (13.1)	10 (4.1)	22.407	0.000*	1
Elementary school	39 (20.5)	36 (14.8)			0.43 (0.18–1.02)
Junior high school	48 (25.3)	59 (24.2)			0.33 (0.14–0.74)
Senior high school	49 (25.8)	67 (27.4)			0.29 (0.13–0.66)
University	29 (15.3)	72 (29.5)			0.16 (0.07–0.38)
Smoking ^a	86 (45.3)	119 (48.8)	0.527	0.509*	0.87 (0.59–1.27)
Hypertension ^a	122 (64.2)	130 (53.3)	5.423	0.024*	1.57 (1.07–2.32)
Diabetes ^a	47 (24.7)	36 (14.8)	6.883	0.010*	1.89 (1.17–3.08)
Prior stroke ^a	57 (30.0)	41 (16.8)	10.641	0.001*	2.12 (1.34–3.35)
Atrial fibrillation ^a	25 (13.2)	18 (7.4)	3.999	0.046*	1.90 (1.00–3.60)
Temperature ($^{\circ}\text{C}$) ^b	36.55 (0.46)	36.5 (0.35)	1.363	0.740+	
Systolic pressure (mmHg) ^b	143.4 (21.9)	140.7 (20.2)	1.924	0.057+	
Diastolic pressure (mmHg) ^b	84.3 (15.2)	85.6 (11.6)	0.346	0.730+	
Glasgow score ^c	14 (13, 15)	15 (14, 15)	5.294	0.000#	
NIHSS score ^c	6 (2, 10)	4 (2, 5)	8.019	0.000#	
OCSP typing ^a					
TACI	12 (6.3)	2 (0.8)	27.735	0.000*	1
PACI	64 (33.7)	46 (18.9)			0.23 (0.05–1.08)
POCI	5 (2.6)	18 (7.4)			0.05 (0.01–0.28)
LACI	109 (57.4)	178 (72.9)			0.10 (0.02–0.46)
Complication ^a	50 (26.3)	23 (9.4)	21.779	0.000*	3.43 (2.01–5.87)
WBC ($\times 10^9 \text{ L}^{-1}$) ^b	7.99 (3.13)	6.54 (2.10)	2.531	0.012+	
Serum glucose (nmol/L) ^c	5.3 (4.5, 7.3)	4.9 (4.3, 6.3)	3.270	0.002#	
Urea nitrogen (mmol/L) ^b	5.97 (2.98)	5.27 (1.64)	2.263	0.024+	
Creatinine ($\mu\text{mol/L}$) ^b	98.9 (34.7)	92.6 (19.1)	2.092	0.037+	
Fibrinogen (g/L) ^b	4.11 (1.4)	3.41 (1.13)	4.086	0.000+	
Time from onset to admission (days) ^c	3 (1, 8)	4 (2, 8)	1.167	0.243#	

* p -Values based on χ^2 -statistic; + p -values based on t -test; # p -values based on Mann–Whitney U -test.^a n (%).^b Mean (S.D.).^c Median (IQR).

or hypertension ($p = 0.024$) or diabetes mellitus ($p = 0.001$) or atrial fibrillation ($p = 0.046$), lower GCS ($p < 0.001$), higher NIHSS ($p < 0.001$), TACI and PACI type ($p < 0.001$), having complication ($p < 0.001$), higher leucocyte count ($p = 0.012$), hyperglycemia ($p = 0.002$), elevated urea nitrogen ($p = 0.024$), creatinine ($p = 0.037$), fibrinogen ($p < 0.001$) and systolic pressure ($p = 0.057$). These variables were then entered into a multivariate logistic regression model. Smoking, temperature, diastolic pressure and time from onset to admission were not significantly different for patients with good and poor stroke recovery.

3.3. Results of the logistic regression model to predict the outcome of ischemic stroke

We categorized the eight statistically significant factors in the univariate analyses for optimum discrimination between good and poor recovery in accordance with those accepted in clinical practice and previous reports. Laboratory data was categorized based on the reference value

of our hospital. The categories obtained were: GCS = 0 (< 11), 1 (≥ 11), NIHSS score = 0 (≤ 3), 1 (4–15), 2 (> 15), WBC = 0 (≤ 10), 1 (> 10), serum glucose = 0 (≤ 5.8), 1 (> 5.8), urea nitrogen = 0 (≤ 7.2), 1 (> 7.2), creatinine = 0 (≤ 125), 1 (> 125), fibrinogen = 0 (≤ 4.0), 1 (> 4.0), systolic pressure = 0 (≤ 140), 1 (> 140). In the logistic regression analysis, sex, history of hypertension or atrial fibrillation or diabetes mellitus, GCS, OCSP typing, complication, systolic pressure, WBC, serum glucose, urea nitrogen, creatinine and fibrinogen were eliminated by forward stepwise variable selection. The poor outcome was associated with old age (OR: 3.505; 95% CI: 2.100–5.849), lower educational level (OR: 0.686; 95% CI: 0.570–0.825), having stroke history (OR: 2.481; 95% CI: 1.442–4.268), and higher NIHSS total score (OR: 2.619; 95% CI: 1.584–4.330) (Table 4). The sensitivity of the model for the prediction of poor recovery was 0.77 and the specificity was 0.71 with an overall probability of being correct of 0.74. The Hosmer and Lemeshow statistic indicated that the goodness fit of the model was good ($p = 0.787$).

Table 4

Results of multivariate logistic regression model to predict the functional outcome of ischemic stroke ($mRS \leq 2$ vs. $mRS > 2$)

Factor	B	S.E.	Wald	Sig	OR	95.0% CI
Constant	0.962	1.200	0.642	0.423		
Age	1.254	0.261	23.039	0.000	3.505	(2.100, 5.849)
Education	−0.377	0.095	15.933	0.000	0.686	(0.570, 0.825)
History of stroke	0.909	0.277	10.784	0.002	2.481	(1.442, 4.268)
NIHSS score	0.963	0.256	14.090	0.001	2.619	(1.584, 4.330)

B: regression coefficient; S.E.: standard error; CI: confidence interval; NIHSS: National Institute of Health Stroke Scale.

4. Discussion

We identified some clinical factors which predict the outcome of ischemic stroke patients in northwest China with hospital-based retrospective study. Our study showed that that age, the educational level, stroke history, and NIHSS score on admission are useful in the prediction of functional outcome of ischemic stroke.

There were many studies concerning the outcome of stroke and there were some conflicting results because of different subjects, different variables, different statistical methods and different end points. The selection of prognostic factors depends on the chosen outcome, as different prognostic factors emerged when predicting functional independence or mortality, respectively. A study of 1754 patients with acute ischemic stroke from the German Stroke Database showed that the prognostic factors for functional independence (Barthel Index ≥ 95 after 100 days) were age, NIH-Stroke Scale at admission, prior stroke and other factors, but educational level was not selected for investigation in this study [15]. In mainland China, there are few studies concerning about functional outcome in ischemic stroke. Furthermore, there is higher incidence of ischemic stroke in northwest area, so we focused on long-term functional outcome of ischemic stroke in the area. To our best knowledge, this is the first study to investigate long-term functional predictors of ischemic stroke in mainland China. As advocated by the principles for evidence-based medicine, we performed a systematic literature search to identify predictors that had been suggested previously. Thus, we were able to consider all these factors simultaneously which is the only means to estimate their relative influence on outcome variable. There is a possibility that some variables were incorrectly documented or were not recorded. In order to minimize this bias, we deliberately selected the variables that clinicians would routinely examine and record. Our resulting models are based on a large data set. These data from the biggest general hospital in the area, 42% patients of those coming from other provinces except Shaanxi are most likely representative of all patients hospitalized for ischemic stroke in five provinces of northwest. In order to avoid the bias of measurement, blinded follow-up was applied and sampling patients were rechecked. The κ -value of two interviewer was 0.847. Although some patients were lost, the lost rate was 11.2% and there were no significant differences in baseline characteristics between the follow-up group and the lost group.

There were 17 factors associated with poor outcome in univariate analysis and 4 factors included old age, lower educational level, having stroke history, higher NIHSS score in multivariate logistic regression. Age has been found to be a significant predictor of mortality and poor outcome in most studies [1,16,17] but not all [18]. Old age was a significant predictor of poor outcome because they had a higher risk of subsequent complications and thereby lower chances to recover from their stroke. Our data confirmed an independent prognostic role of increasing age on poor functional outcome.

There are few studies about the level of education as a predictor of stroke but results were conflicting. Some studies showed there is no relation between educational level and mortality [19,20], but some studies got the opposite conclusion [21]. The quality of life in stroke survivors was also related to education level [22,23]. They did not get the same results because of applying different end points and different methods. Our study showed that lower education level was associated with poor outcome. There are two possible reasons. One is that patients with high education are so well-informed that they have abundant knowledge of the disease and its nursing. Another is that patients with high education often obtain better economic conditions, compare with those with low education. Moreover, economic conditions are significantly correlated with fatality, irrespective of gender, age, the severity of ischemic strokes.

Stroke is the leading cause of chronic disability. At least half of survivors have disability to a certain degree. Those who had stroke history were more likely to recur [24]. Some studies only included the patients with a first-ever stroke, but prior stroke has been proposed as a predictor for mortality and functional dependence [25–27]. The patients having stroke history were more likely becoming poor outcome compared with those without stroke history in the present study.

Initial severity of stroke has been shown to be a predictor of functional outcome in many studies. Almost all prognostic models included the scale that measure of severity of stroke [28,29]. There are many stroke scales used in stroke studies as a measure of severity of stroke such as the National Institute of Health Stroke Scale (NIHSS), the Scandinavian stroke scale (SSS), the European stroke scale (ESS), the Canadian stroke scale (CSS). The NIHSS is one of the most reliable and valid instruments of clinical measurement in stroke and is most extensively used. NIHSS measures an overall degree of neurologic impairment involving eval-

uations of level of consciousness, gaze, visual fields, facial movement, motor function, limb ataxia, sensory, best language, dysarthria, neglect. These parameters not only reflect the severity of the patients' neurological deficit [30], but also provide powerful prognostic information and form the major component of the prediction algorithm [31–33]. In our analyses, the total NIHSS score at admission was a prognostic factor for outcome of ischemic stroke.

A limitation of this study is its retrospective design. The variables were those recorded in the medical records rather than being prospectively and selectively collected. Some factors such as TOAST type and imaging variables were not taken into consideration, as without explicit prospective definitions they cannot be reliably drawn from the charts. While this means that standardization of the study factors was not possible, it suggests that the results may be more likely to reflect actual clinical practice. Secondly, we defined a number of risk factors such as smoking and hypertension as binary variables although duration and severity are important. Finally, we only included patients admitted to our hospital, which might be in relatively better economic conditions. Although there were some biases in case selection, the results of the present study were suggestive and helpful at least in managing hospital-admitted patients.

5. Conclusion

In conclusion, with a retrospective method, we have identified some factors which predict the outcome of ischemic stroke patients in northwest China using simple and easily identified clinical features. Our data confirmed that old age, lower educational level, having stroke history, higher NIHSS score were the independent predictor of poor outcome. Next, we will assess the validation and reliability of model with prospective study before it can be incorporated into routine clinical practice.

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